

QUESTION 1: (12 Marks) [**START A NEW PAGE**]

(a) Solve the equation $5(x-2) - 6 = 29$.

(b) Simplify $\frac{1}{x-2} - \frac{1}{x}$.

(c) Find $\log_5 12$ correct to 2 decimal places.

(d) Factorise fully (a) $9y^2 - 36$.

(b) $2t^2 - 7t + 6$.

(e) Given that p and q are rational, find the value of p if $2\sqrt{3}(4+\sqrt{3}) = p + q\sqrt{3}$.

QUESTION 2: (12 Marks) [**START A NEW PAGE**]

The co-ordinates of the points A, B and C are $(-4,3)$, $(0,5)$ and $(9,2)$ respectively.

(a) Find the length of the interval BC.

(b) Show that the equation of the line l , drawn through A and parallel to BC is $x + 3y - 5 = 0$.

(c) Find the co-ordinates of D, the point where the line l meets the x-axis.

(d) Prove that ABCD is a parallelogram.

(e) Find the perpendicular distance from the point B to the line l .

(f) Hence or otherwise find the area of parallelogram ABCD.

QUESTION 3: (12 Marks) [**START A NEW PAGE**]

(a) Find the derivatives, leaving your answer in simplest form:

(i) $4x\sqrt{x}$.

(ii) $x \tan x$.

(iii) $\frac{4x-1}{2x+3}$.

(b) Find, leaving your answer in exact form:

(i) $\int_0^4 \frac{1}{3x+2} dx$.

(ii) $\int_4^9 \frac{6}{x\sqrt{x}} dx$

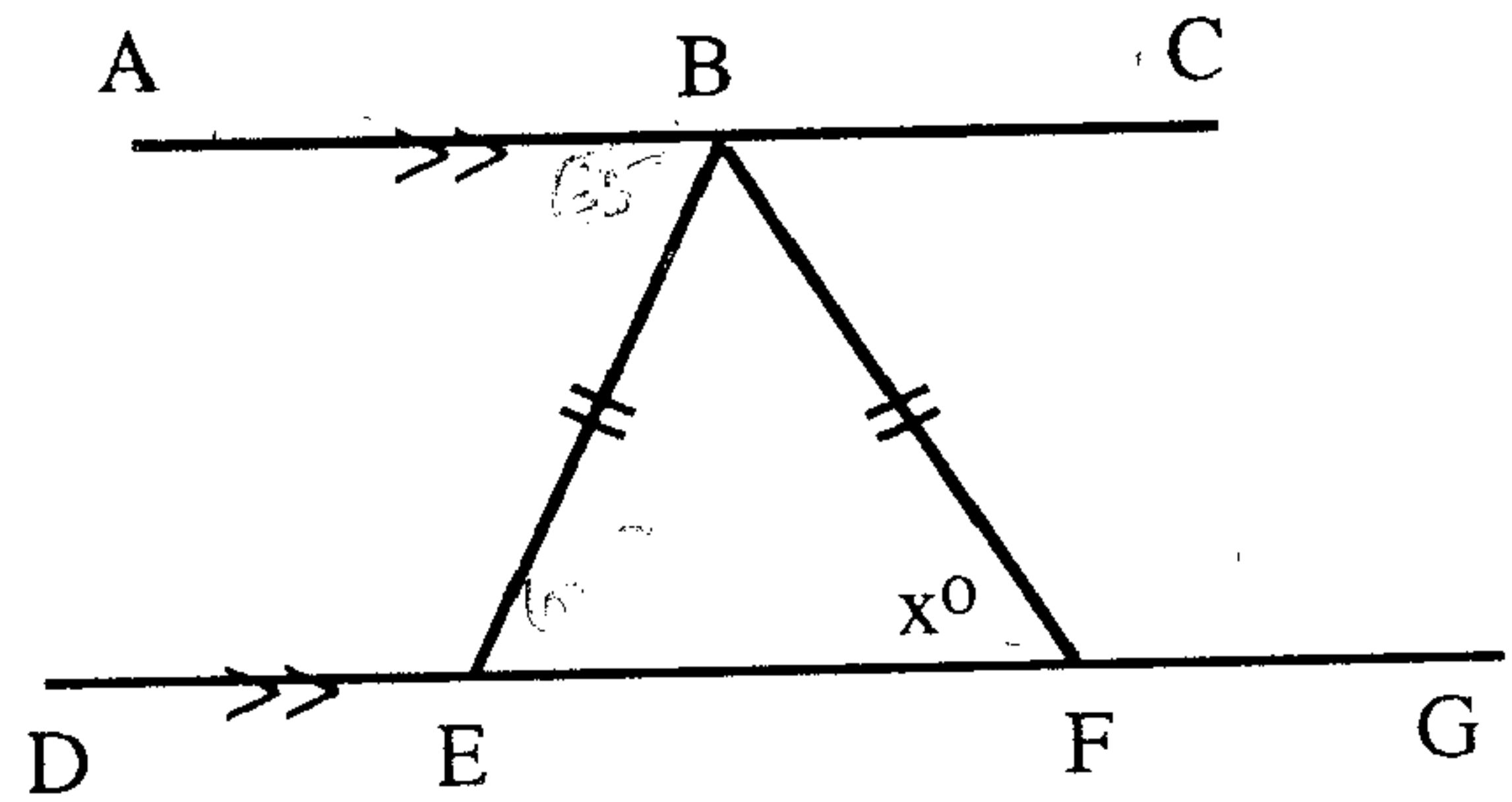
(c) A curve has gradient function $\frac{dy}{dx} = e^{3x}$. Find the equation of the curve if it passes through the point (0,2).**QUESTION 4:** (12 Marks) [**START A NEW PAGE**]For the parabola $y = \frac{1}{8}x^2 - x + 3$,

(a) Find the co-ordinates of the vertex and focus.

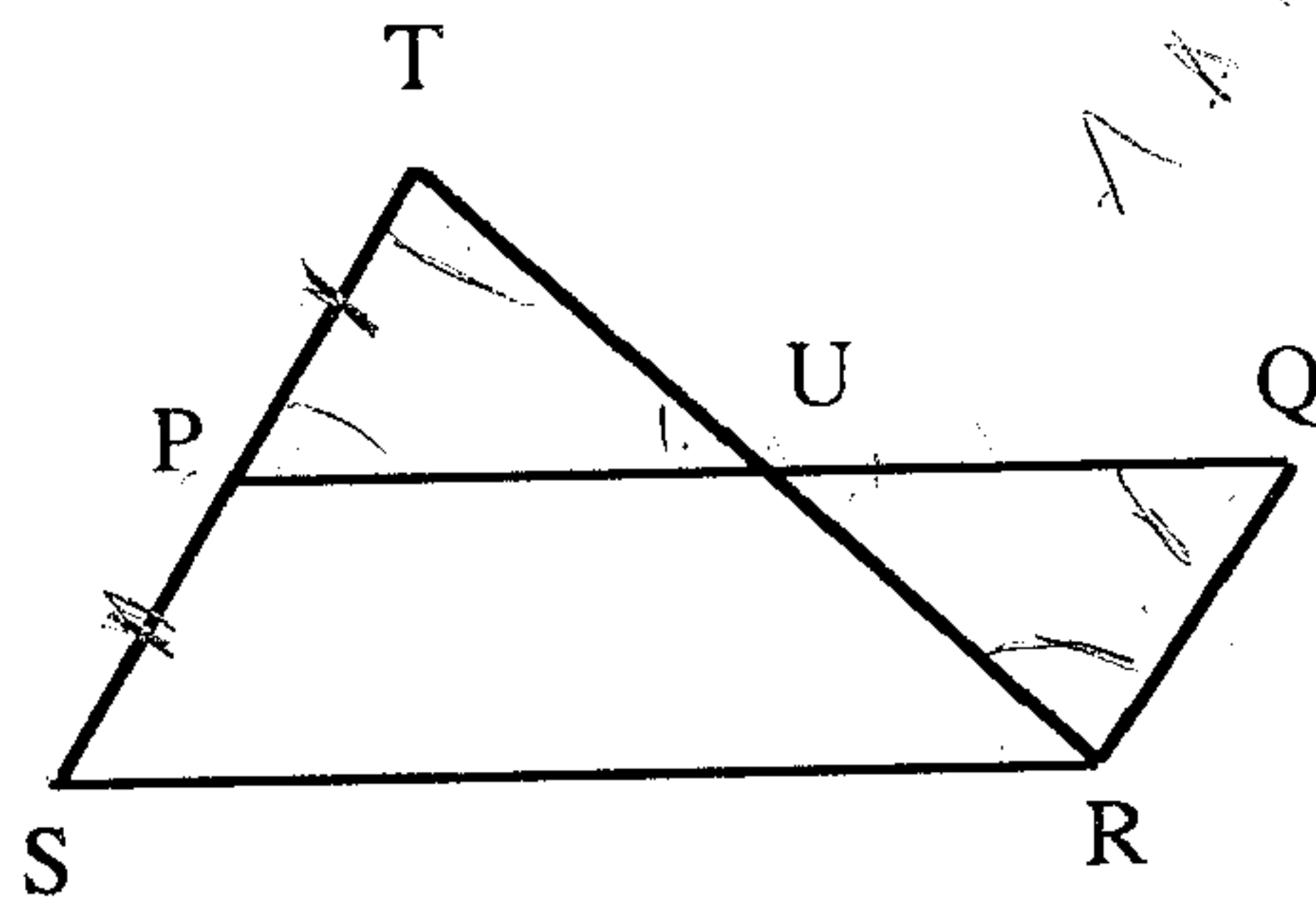
(b) Find the equation of the normal to the parabola at the point where $x = -4$. Write your answer in general form.(c) Find the point on the parabola at which the tangent is parallel to $y = 3x + 1$.

QUESTION 5: (12 Marks) [**START A NEW PAGE**]

- (a) $AC \parallel DG$, $\angle ABE = 65^\circ$, $BE = BF$.
Copy the diagram to your examination paper.
Find the value of x , giving all reasons.



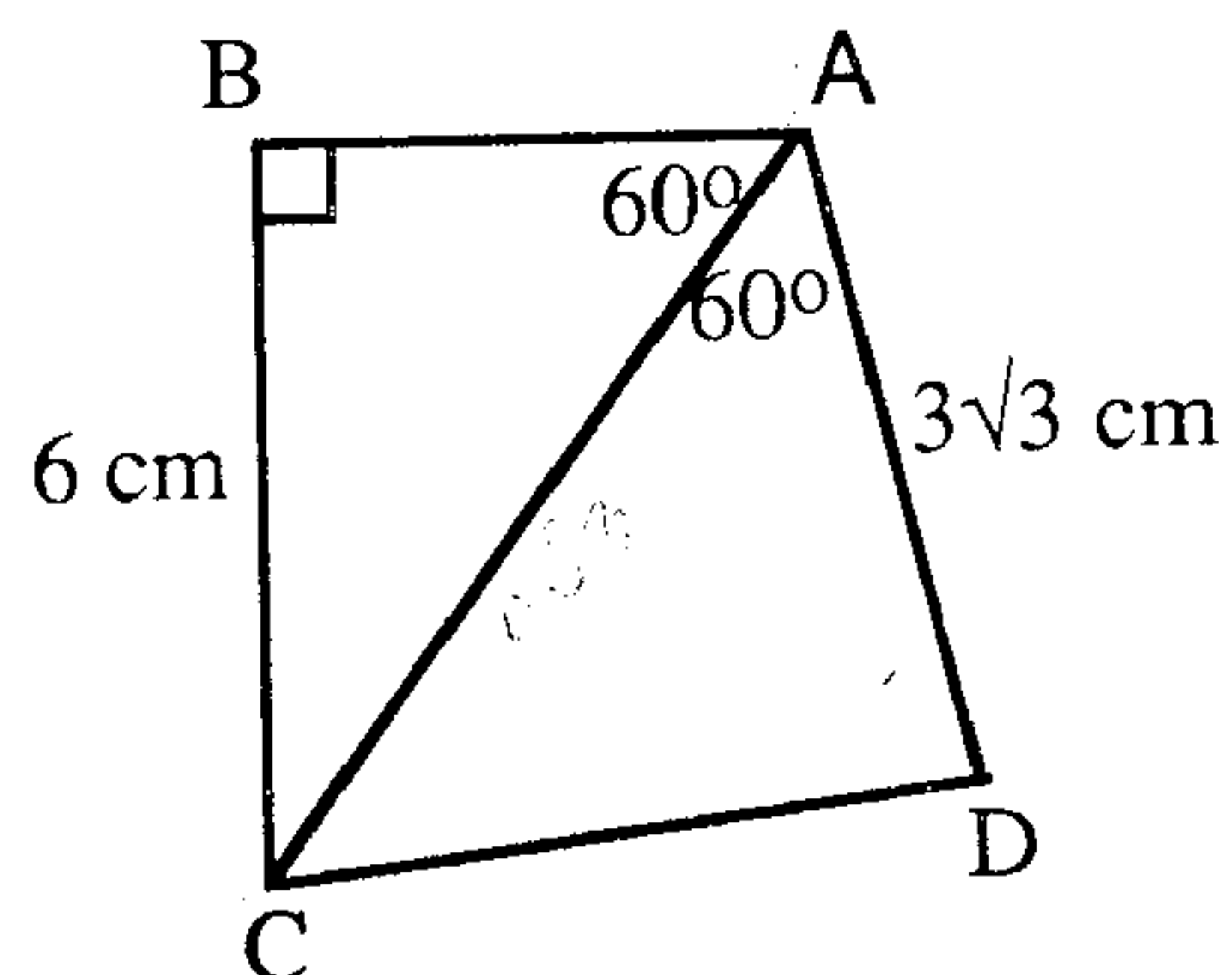
- (b) PQRS is a parallelogram and SP is extended to T so that $SP = PT$.
Copy the diagram to your examination paper.
Using congruent triangles or otherwise prove that U is the midpoint of PQ.



- (c) In the diagram, not drawn to scale,
 $\angle BAC = \angle CAD = 60^\circ$,
 $\angle ABC = 90^\circ$, $BC = 6$ cm and
 $AD = 3\sqrt{3}$ cm.

Find as exact values:

- (i) the length of AC,
(ii) the length of CD.

**QUESTION 6:** (12 Marks) [**START A NEW PAGE**]

- (a) The third term of an arithmetic progression is 42 and the seventh term is 10.

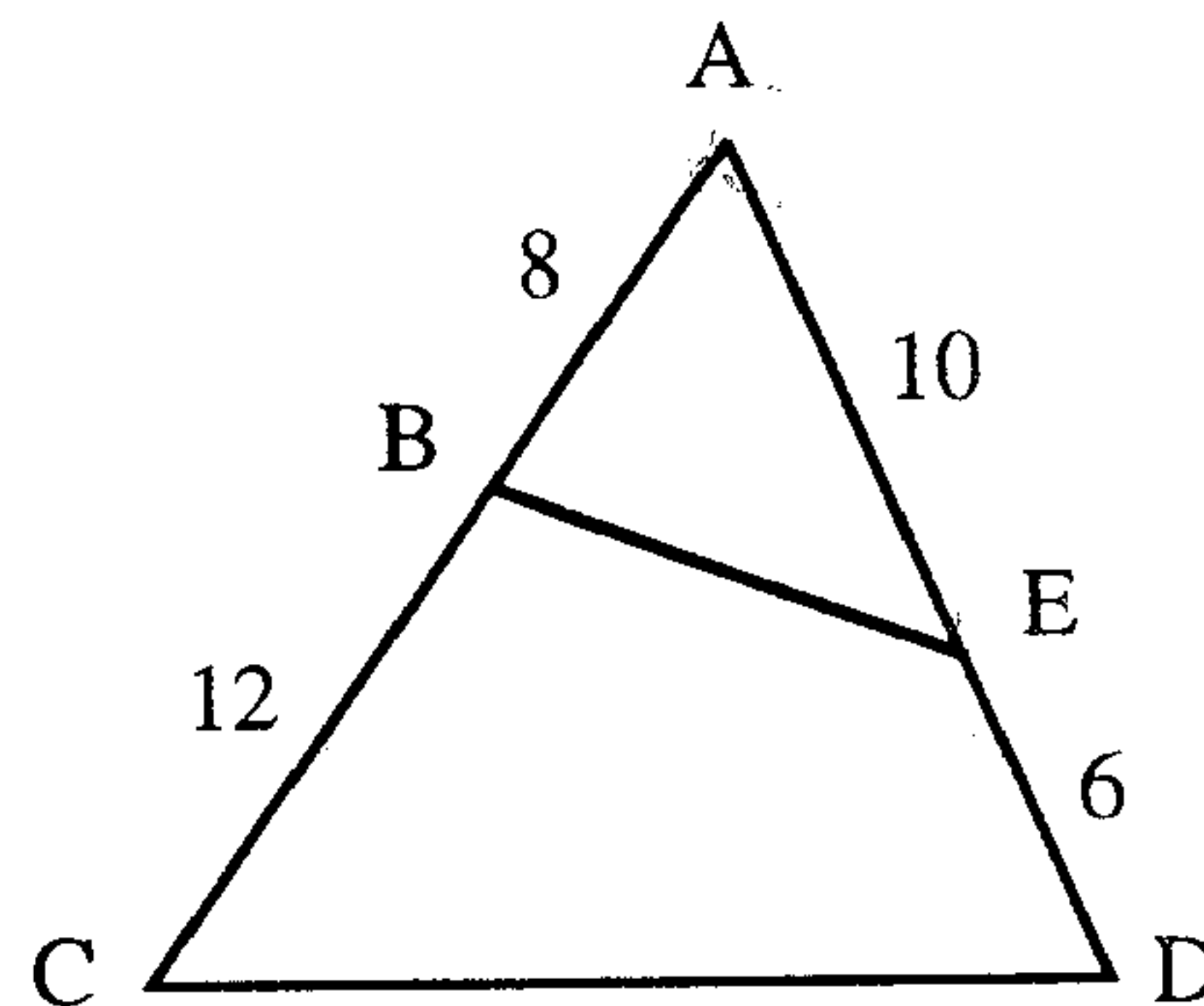
- (i) Find the common difference and the first term.
(ii) Find the sum of the first twenty terms.

- (b) Ray deposits \$50 into a superannuation fund at the start of each month. The fund pays 15% p.a. interest which is compounded at the end of each month.

- (i) Find the value of the fund at the end of 10 years.
(ii) How many months will Ray have to contribute to the fund if he wishes the fund to be worth \$25 000?

QUESTION 7: (12 Marks) [**START A NEW PAGE**]

- (a) (i) Prove that $\triangle ABE$ and $\triangle ACD$ are similar.
- (ii) Given that $CD = 18$, find the length of BE .
Give all reasons.



- (b) A train runs between two stations stopping at each. Its velocity v km/min, t minutes after leaving the first station, is given by $v = 0.12t(6-t)$.

Find

- (i) the time taken to travel between the two stations.
- (ii) the maximum positive velocity attained.
- (iii) the distance between the two stations.

QUESTION 8: (12 Marks) [**START A NEW PAGE**]

- (a) (i) Sketch the curve $y = 4 - x^2$, clearly showing its intercepts with the co-ordinate axes.
- (ii) Find the volume of the solid formed when the area bounded by the curve $y = 4 - x^2$ and the x -axis is rotated one complete revolution about the y -axis.
- (b) For the curve $y = x^4 - 8x^3 + 18x^2 - 90$ find the co-ordinates of any stationary points and determine their nature.

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2/3 Unit. 1992

1. (a) $x = 9$
(b) $\frac{2}{x(x-2)}$

(c) 1.54

(d) $9(y-2)(y+2)$

(e) $p = 6$

2. (a) $BC = 3\sqrt{10}$

(b) $m_{BC} = -\frac{1}{3}$

$x + 3y - 5 = 0$

(c) $D(5, 0)$

(d) $m_{AC} = m_{BD} = \frac{1}{2}$

$\therefore ABCD$ para.

(e) $\sqrt{10}$

(f) $A = 30 \text{ V}^2$

3. (a)(i) $6\sqrt{x}$

(ii) $x \sec^2 x + \tan x$

(iii) $\frac{14}{(2x+3)^2}$

(b)(i) $\frac{1}{3} \ln 7$

(ii) 2

(c) $y = \frac{1}{3} e^{3x} + \frac{5}{3}$

4. (a) Focal length 2

Vertex $(4, 1)$

Focus $(4, 3)$

(b) $x - 2y + 22 = 0$

(c) $(16, 19)$

5. (a) $x = 65$

(b) $\Delta PTU \cong \Delta GRU$ (AAS)

$PV = UR \therefore U$ mid point.

(c)(i) $AC = 4\sqrt{3}$

(ii) $CD = \sqrt{39}$

6. (a)(i) $a = 58$

(ii) $S_{20} = -360$

(b)(i) $\$13,932.86$

(ii) 159 months.

7. (a)(i) $AC:AE = AD:AB = 2:1$
and $\hat{A}$ common

(ii) $BE = 9$

(b)(i) 6 minutes.

(ii) $v_{\max} = 1.08 \text{ km/min}$

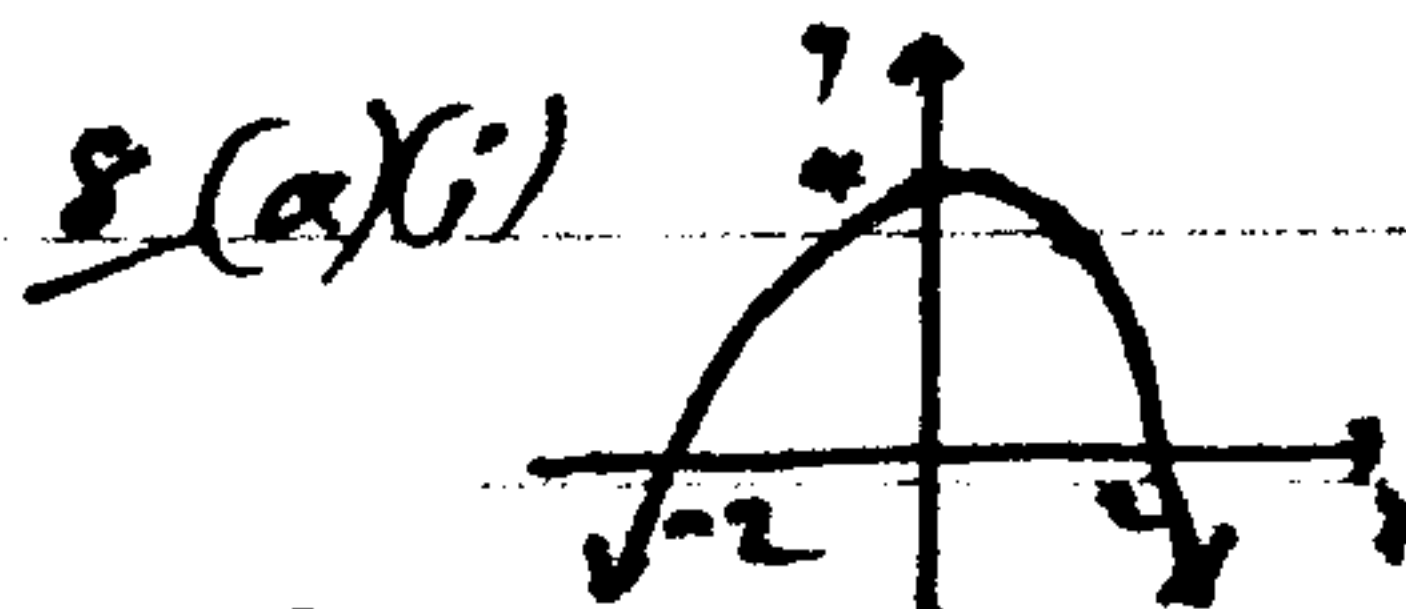
(iii) 4.32 km.

10. (a) $c < -2\frac{1}{4}$

$\Rightarrow \Delta < 0$ or $a > 0$

(b)(i) $l = \frac{2}{r} - r$

(ii) $V = \frac{2\pi r}{3} \sqrt{1-r^2}$



(ii) $V = 8\pi \text{ V}^3$

(b) Min T.P. at $(0, -90)$

Horizontal point of inflection at $(3, -63)$

9. (a)(i) $P(66) = \frac{3}{10}$

(ii) $P(>12\text{ hrs}) = \frac{7}{10}$

(iii) $P(24) = \frac{1}{3}$

(b)(i) $t = 20$ minutes.

(ii) $V = \frac{200}{\pi} (1 - \cos \frac{\pi t}{20})$

(iii) $V_{\max} = \frac{400}{\pi} \text{ L.}$

(iii) Absolute max. occurs

when $r = \frac{1}{\sqrt{2}}$.